Curated Research Articles

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- **A Closed-Loop Direct Recycling Process for Prussian White From Scrap and End-of-Life Sodium-Ion Batteries** — score: 1.000 This article presents a closed-loop direct recycling method for Prussian White cathodes from both manufacturing scrap and end-of-life sodium-ion batteries. Using an innovative low-temperature aqueous resodiation process, the researchers effectively restore essential properties of the cathodes, achieving nearly pristine capacity and significant cycling stability while drastically reducing energy costs compared to conventional synthesis methods. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Mn/V Co-Doping Enables Multielectron Transfer and Above-Theoretical Capacity in Na4Fe3(PO4)2P2O7 Cathode** — score: 1.000 The article discusses the development of a high-performance Na3.5Fe2Mn0.5V0.5(PO4)2P2O7 composite cathode for sodium-ion batteries, achieved through synergistic Mn/V co-doping and carbon modification. This cathode surpasses the theoretical capacity limit, recording an initial discharge capacity of 149.11 mAh g−1 and showcasing improved energy density, rate capability, and cycling stability, thus paving the way for enhanced polyanionic material utilization in battery applications. Journal: *Wiley: Small Methods: Table of Contents*
- **Linking DSC/TGA to Cell Levels: Energetics, Evolved Gases, and Thermal Safety of NMC811-Graphite Micro-Cell** — score: 1.000 This study presents a component-resolved framework that links differential scanning calorimetry (DSC), thermogravimetric analysis (TGA), and evolved gas analysis to investigate the thermal behavior of NMC811-Graphite micro-cells. The research highlights how the presence of a separator influences reaction pathways and reduces energy release in micro-cells compared to separator-free configurations, providing a foundation for safety assessments and thermodynamic modeling in battery systems. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **The critical role of water structure in anion transport** — score: 1.000 The article discusses the interplay between water structure and anion transport, revealing that ionic mobility in confined environments is influenced by the collective dynamics of ions, water, and polymers. Utilizing neutron spectroscopy, total scattering, and molecular dynamics simulations, the research highlights key structural and transport properties that underpin ionic conductivity. Journal: *RSC - Energy Environ. Sci. latest articles*
- **Spatial shift of Li nucleation onto electrically conductive SEI residues** — score: 0.900 The study reveals that residual solid electrolyte interphase (rSEI) layers in batteries are not inert but become conductive over time, leading lithium nucleation to shift from the current collector to the rSEI surface. This shift increases the likelihood of electrical disconnection and contributes significantly to battery degradation and failure. Journal: *Joule*
- **Iron-Based Sodium-Salt Composite Cathode for Sodium-Ion Batteries** — score: 0.900 This study presents a novel sodium-ion battery cathode made from a composite of Na2SO4, iron powder, and CuS, synthesized through mechanochemical means. The composite enables effective multi-electron transfer in iron redox chemistry, achieving a reversible capacity of 149 mAh g−1 at approximately 2.6 V, highlighting the potential for high-performance, cost-effective energy storage solutions using abundant materials. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Dehydrating A Borate Additive Unlocks Fast-Charging Graphite** — score: 0.900 The study introduces an anhydrous lithium bis(catecholato)borate (LiBCB) additive that significantly enhances the performance of graphite anodes in lithium-ion batteries by forming a thin, effective solid electrolyte interphase (SEI) and mitigating issues related to interfacial polarization and lithium plating. Through a one-step, non-aqueous synthesis that eliminates water, LiBCB demonstrated superior cycling stability and capacity retention compared to traditional additives like LiBOB, highlighting its potential as a promising alternative for fast-charging applications. Journal: *Wiley: Advanced Energy Materials: Table of Contents*

- **Hydrogel-Engineered Interface Suppressing Water Activity by Hydrogen Bonds and Regulated Solvation for Stable Zn Metal Anodes** — score: 0.800 This article presents a novel approach to enhancing the stability and longevity of zinc metal anodes in aqueous Zn-ion batteries by applying an in situ hydrogel layer. This hydrogel, composed of polyacrylamide and polyethylene glycol, effectively suppresses water activity and modifies Zn2+ solvation through strong hydrogen bonding and ion exchange, resulting in remarkable cycling durability exceeding 6600 hours and improved electrochemical performance in full cells. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Practical Perspectives and Energy Density Criteria for Current Collector Engineering in Anode-Free Li-Metal Batteries** — score: 0.800 The article reviews engineering strategies for current collectors in anode-free Li-metal batteries, focusing on how various modifications affect energy metrics while addressing critical issues like mass and volume penalties. It introduces a "Thin-Li benchmark" to evaluate practical gains from engineering enhancements and identifies key constraints and failure mechanisms that limit battery performance, ultimately proposing directions for future research to optimize current collector designs and improve energy density. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Boosting Proton Conduction and Oxygen Reduction Kinetics via In Situ Reverse Atom Capture for Protonic Ceramic Fuel Cell Cathodes** — score: 0.800 The article presents a novel reverse atom capture technique used to enhance the performance of protonic ceramic fuel cell (PCFC) cathodes by constructing a BSS@PBSCF heterostructure in situ. This method increases oxygen vacancy concentration and accelerates proton transport, achieving a peak power density of 1.90 W cm−2 at 650°C with excellent stability over 210 hours, thus advancing the commercialization prospects of PCFCs. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Boosting Oxygen Reduction via Nanotwinned High-Entropy Aerogels With Deficient Pd Coordination and Multi-d-Orbital Hybridization for Zinc-Air Battery Applications** — score: 0.800 The study presents high-entropy alloy aerogels (HEAAs) made from PdPtCuNiCo, showcasing a unique hierarchical porous structure that enhances mass transport and promotes efficient oxygen reduction reaction kinetics in alkaline electrolytes. The combination of atomic-scale defect engineering, including nanotwins and unsaturated Pd sites, along with multi-d-orbital electron transfer, results in superior electrocatalytic performance for zinc-air battery applications, achieving remarkable stability and current density compared to conventional catalysts. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Leveraging the Amorphous Nature of Sn–P Alloys for Improved Stability and Energy Density in Na-Ion Batteries** — score: 0.800 The study evaluates Sn-P alloys as anode materials for sodium-ion batteries, highlighting that amorphous SnP3 exhibits superior cycling stability and volumetric capacity compared to the crystalline Sn4P3, which experiences rapid capacity decay due to phase separation during sodiation. The findings underscore the advantages of the amorphous structure in enhancing the performance of Na-alloy anodes, making amorphous SnP3 a promising candidate for high-energy-density applications. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Ultrathin, Tough Asymmetric MOF-Based Separator Enabling Rapid Selective Ion Transport and Stable Aqueous Zn-Ion Batteries** — score: 0.800 The article presents a novel ultrathin asymmetric separator, created by in situ growing a microporous Ni3(HITP)2 layer on a bacterial cellulose membrane, which effectively enhances zinc ion transport and minimizes dendrite formation in aqueous Zn-ion batteries. This innovative design leads to improved electrochemical stability and performance, marking a significant advancement in the development of high-efficiency zinc-based energy storage systems. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Enabling high-voltage and wide-temperature quasi-solid-state batteries via spatially engineered polyether electrolyte interfaces** — score: 0.800 The researchers develop a quasi-solid-state polymer electrolyte that features spatially engineered interfaces to enhance the stability and safety of lithium metal batteries. This innovative design allows the batteries to function effectively across a broad temperature range and with various cathode materials, paving the way for safer, high-energy rechargeable battery technologies. Journal: *Joule*

- **Organic Cocrystal Interface Engineering for Stable Aqueous Zinc-Ion Batteries** — score: 0.800 This study introduces organic cocrystals formed from melamine and isophthalic acid to enhance zinc anodes in aqueous zinc-ion batteries by creating a stable interfacial layer. This innovative approach improves ion transport and suppresses harmful side reactions, resulting in outstanding cycling stability, with test cells showing over 4200 hours of reversible operation and significant capacity retention after extensive cycling. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Single Metal Atoms for Energy Storage and Conversion: Recent Advances, Challenges, and Future Perspectives** — score: 0.800 The article provides an in-depth review of recent advancements in single-atom catalysts (SACs) used for energy storage and conversion, examining their synthesis, structural characteristics, and functional mechanisms. It discusses various preparation methods and support materials, highlights their applications in electrocatalysis and energy devices, and addresses challenges like atom stability and large-scale production while offering insights into future research directions for optimizing SACs in sustainable energy technologies. Journal: *Wiley: Small Methods: Table of Contents*
- **Managing silicon burn-out via onboard material diagnostics for durable high-energy density batteries** — score: 0.700 The study by Wan et al. addresses the challenge of silicon aging in graphite anodes, which limits battery capacity. By utilizing routine charging data, the researchers identify the state-of-charge window for silicon usage and propose a method for battery management systems to implement age-informed thermal controls that better protect high-energy materials, thus optimizing capacity and longevity. Journal: *Joule*
- **Contact loss in all-solid-state Li-ion batteries via deposition of impurities** — score: 0.700 The study reveals that interfacial contact loss in all-solid-state lithium-ion batteries is significantly influenced by the accumulation of insulating impurities within the lithium anode, which form a porous layer at the Li-electrolyte interface during the stripping process. This layer hinders lithium flux, contributing to battery performance degradation. Journal: *Joule*
- **Machine learning-assisted materials design, mechanistic insights, and predictive modeling for high performance lithium-sulfur batteries** — score: 0.600 This article discusses advancements in the design and optimization of lithium-sulfur batteries using machine learning techniques. The authors explore how these methods provide mechanistic insights and enhance predictive modeling, potentially leading to improved battery performance and efficiency. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Monolayer MoS2 Enabled by Intercalation-Confinement Synergy Toward Ultrafast and Stable Sodium Storage** — score: 0.600 The article presents a novel method for synthesizing monolayer MoS2 nanosheets encapsulated in nitrogen-doped carbon shells, enhancing their performance as anodes in sodium-ion batteries. Utilizing an intercalation-confinement synergy, the resulting hollow structures demonstrate improved electrical conductivity, optimized sodium ion dynamics, and remarkable cycling stability, achieving an exceptional energy capacity and longevity suitable for practical applications. Journal: *Wiley: Small Methods: Table of Contents*
- **Strong electron push–pull effect and low transition barrier boosting non-metallic COFs photoelectrode for ultrahigh-rate photo-assisted Li–air batteries** — score: 0.600 The article discusses the development of donor-acceptor type covalent organic frameworks (COFs) that feature a strong electron push-pull effect and a low transition barrier, which significantly enhance the performance of photo-assisted lithium-air batteries. These COFs demonstrate improved round-trip efficiency, high rate capability, and durability, making them promising materials for energy storage applications. Journal: *RSC - Energy Environ. Sci. latest articles*
- **3D Interconnected Zn-Ion Pathways Through Anionic Nanochannels Surrounding Bacterial Cellulose Nanofibers by Repetitive Interfacial Assembly for Aqueous Zn-Ion Battery Separators (Adv. Energy Mater. 24/2026)** — score: 0.600 The authors present a novel separator made from bacterial cellulose nanofibers, featuring interconnected anionic nanochannels created through repetitive interfacial assembly. This design facilitates efficient Zn2+ transport and helps to

prevent dendrite formation in aqueous Zn-ion batteries, resulting in improved electrochemical performance. Journal: *Wiley: Advanced Energy Materials: Table of Contents*

- **Mechanistic Pathways of Solid Electrolyte-Dependent Degradation in Ni-rich Cathodes for All-Solid-State Batteries** — score: 0.500 The article explores the specific degradation mechanisms of Ni-rich cathodes in all-solid-state batteries, focusing on how variations in solid electrolyte characteristics influence these pathways. The findings aim to enhance the understanding of battery performance and longevity by identifying critical factors underlying the degradation process. Journal: *ScienceDirect Publication: Energy Storage Materials*

- **Design Principles, Interfacial Regulation, and AI-Guided Opportunities of Hydrogel Electrolytes for Zinc Metal Batteries in Extreme Conditions** — score: 0.400 The article reviews the design principles and interfacial regulation of hydrogel electrolytes in zinc metal batteries (ZMBs), emphasizing their potential for safe and flexible energy storage. However, it highlights challenges related to stability under extreme conditions and discusses AI-guided strategies to enhance their performance and application in real-world scenarios. Journal: *RSC - Energy Environ. Sci. latest articles*

- **Flowing zinc slurry for long-duration energy storage** — score: 0.400 The article discusses the development of a novel ligand-confined, flowable zinc slurry for aqueous zinc batteries, addressing the challenges of limited reversibility and enhancing their durability and efficiency for long-duration energy storage applications. This innovation aims to make zinc-based energy storage solutions more practical and accessible. Journal: *Nature Energy*

- **Simultaneous Prediction of Three Key Li-ion battery Indicators through Multi-Task Learning** — score: 0.400 The article presents a multi-task learning approach aimed at simultaneously predicting three critical indicators of lithium-ion battery performance: State of Health (SoH), State of Charge (SoC), and State of Function (SoF). The authors emphasize the importance of accurate battery state diagnostics for operational safety and assessing the viability of batteries for second-life applications. Journal: *RSC - EES Batteries latest articles*

- **Opportunities and challenges for the expansion of LFP battery supply chains** — score: 0.300 The article discusses the potential for lithium iron phosphate (LFP) battery supply chain expansion, emphasizing the need for better impurity characterization and the identification of cost-effective iron precursors. It highlights a possible conflict in resource allocation, as the increasing demand for phosphorus for LFP batteries could significantly impact the availability of phosphate rock used for fertilizers. Journal: *RSC - EES Batteries latest articles*

- **Lighting the way to improve battery safety with sensing innovations** — score: 0.200 The article explores recent advancements in diagnostic technologies for commercial batteries aimed at enhancing their reliability and safety. By enabling earlier detection of failures, these innovations can improve battery performance and foster consumer confidence in energy storage solutions. Journal: *Joule*

- **Wide-Temperature Metal-Sulfur Batteries: Challenges and Opportunities** — score: 0.200 The article discusses the potential of metal-sulfur batteries for operation across diverse temperature ranges, highlighting the challenges these systems face, such as stability and performance issues, while also identifying new opportunities for advancements in energy storage technology. The work emphasizes the importance of addressing these challenges to harness the full potential of metal-sulfur batteries in various applications. Journal: *ScienceDirect Publication: Energy Storage Materials*

- **Scaling vanadium flow battery for long-duration energy storage** — score: 0.200 The article addresses the challenges faced by vanadium flow batteries in transitioning from pilot projects to large-scale implementation due to existing bottlenecks. Researchers Xianfeng Li and colleagues explore strategies to enhance the reliability and cost-effectiveness of these batteries for long-duration energy storage applications in the multi-hundred-megawatt range. Journal: *Nature Energy*

- **Crosstalk Effect of Covalent Bonds Reinforces Structural and Thermal Stability of Li-Rich Mn-Based Layered Cathodes** — score: 0.200 The article discusses how the presence of crosstalk effects due to covalent bonds enhances both the structural integrity and thermal stability of lithium-rich manganese-based layered cathodes. This research highlights the importance of covalent interactions in improving the performance of battery materials, which is crucial for advancing energy storage technologies. Journal: *ScienceDirect Publication: Energy Storage Materials*

- **Structure Dependent Optimization of Pyrrolidinium Bromide Complexing Agents for High Energy Density Zinc–Bromine Flow Batteries** — score: 0.200 The article investigates the optimization of pyrrolidinium bromide complexing agents to enhance the performance of zinc–bromine flow batteries. The authors focus on how structural variations in these agents can significantly influence the energy density and overall efficiency of the battery systems. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Tailoring Key Intermediates via Au-Co Dual Sites for Urea Electrosynthesis From Co-Reduction of CO2 and Nitrate** — score: 0.200 The study presents a dual-site catalyst, Au24Cd/CZ9-700, which effectively enhances urea production through the electrochemical coupling of CO2 and NO3− by spatially separating the generation of key intermediates on gold and cobalt sites. This selective approach allows for efficient C–N bond formation and achieves a high urea production rate and Faradaic efficiency, demonstrating a novel strategy for improving electrocatalytic systems in sustainable chemical synthesis. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Hydrogen-free exsolution of Ir–Fe nanoalloys on the surface of solid oxide cell perovskite air electrodes** — score: 0.200 The article discusses a novel approach for forming Ir–Fe nanoalloys on perovskite air electrodes without the use of hydrogen, driven by the affinity between the metals. This hydrogen-free exsolution method could enhance the performance and stability of solid oxide cells by improving the surface characteristics of the electrodes. Journal: *RSC - Energy Environ. Sci. latest articles*
- **Prospects for mining critical materials from nontraditional water sources to meet decarbonized energy needs** — score: 0.100 The article evaluates the potential of nontraditional water sources, like produced water and acid mine drainage, to extract critical materials such as magnesium and lithium essential for decarbonized energy technologies. It highlights the need for diversification in supply chains and offers insights into the technological and geographic factors that could inform future research efforts in material recovery. Journal: *Joule*
- **Author Correction: Demand-side innovation is the priority for decarbonizing materials** — score: 0.000 The article corrects information regarding the significance of demand-side innovation as a critical approach to achieving decarbonization in material sectors. It emphasizes that focusing on how materials are utilized and the demand for sustainable practices is essential for effective climate action. Journal: *Nature Reviews Materials*
- **C-glycoside synthesis via radical cross-coupling of glycohydrazides** — score: 0.000 This article discusses a novel synthetic approach for C-glycosides through the radical cross-coupling of glycohydrazides, providing insights into the reaction mechanisms and potential applications in carbohydrate chemistry. The study presents a significant advancement in the field by offering a more efficient method for producing C-glycosides, which are valuable in medicinal chemistry. Journal: *Nature*
- **Stereoretentive decarbonylative C(sp3)-C(sp3) cross-coupling** — score: 0.000 The article discusses a novel method for stereoretentive decarbonylative $C(sp^3)$-$C(sp^3)$ cross-coupling, offering insights into the efficiency and versatility of this chemical process. This technique enhances the ability to construct complex carbon frameworks while maintaining the stereochemical integrity of the reactants. Journal: *Nature*
- **Isotopic evidence for a cold and distant origin of 3I/ATLAS** — score: 0.000 The article presents isotopic analysis indicating that the comet 3I/ATLAS originated from a cold, distant region, suggesting that its formation occurred far from the sun. This research enhances our understanding of the comet's composition and the conditions prevalent in its early environment, shedding light on the dynamics of celestial bodies in the outer solar system. Journal: *Nature*

- **A spacecraft is falling to its doom — can NASA save it?** — score: 0.000 The article discusses a potential rescue mission for the Swift space observatory, which is currently in peril. Successful intervention efforts could offer hope for similar preservation strategies for the Hubble Space Telescope's future. Journal: *Nature*
- **The first ticking 'nuclear clocks' are here — what can they do?** — score: 0.000 Two research teams have successfully developed advanced "nuclear clocks," representing a significant advancement in precision timekeeping. These innovative clocks exploit the predictability of nuclear transitions to achieve unprecedented accuracy, potentially transforming fields such as navigation and fundamental physics. Journal: *Nature*
- **Will AI spark a scientific renaissance — or a diffuse monoculture?** — score: 0.000 The article explores the potential of artificial intelligence to enhance scientific research, highlighting that its success will hinge on a commitment from the scientific community to prioritize innovative ideas over the quick results that AI can provide. The interaction between AI capabilities and the values of researchers, funders, and reviewers will ultimately shape the future of scientific inquiry. Journal: *Nature*
- **Forty years of high-temperature superconductivity** — score: 0.000 The article reflects on four decades since the discovery of superconductivity at 35 kelvin, which sparked extensive research into high-temperature superconductors. It highlights the ongoing challenges and mysteries surrounding this unusual state of matter, which continue to drive scientific inquiry and innovation. Journal: *Nature*
- **Make science more reliable: study people as they go about their lives** — score: 0.000 The article addresses the pressing issue of generalizability in behavioral sciences, highlighting the need for research methodologies that reflect real-world conditions rather than relying solely on traditional lab settings. It builds on the ongoing conversation surrounding the replication crisis, emphasizing that improving the reliability of scientific findings requires studying individuals in their everyday environments. Journal: *Nature*
- **Cancer cells adopt unprecedented strategies to produce a molecule that protects them from iron-dependent death** — score: 0.000 Recent research reveals that cancer cells utilize unique mechanisms to produce spermine, which binds to iron and counteracts ferroptosis, a form of cell death. This discovery could lead to novel therapeutic approaches for managing tissue damage and cancer treatment. Journal: *Nature*
- **Why heritage sites are at risk in a warming world — and how to save them** — score: 0.000 The article discusses the increasing risks that climate change poses to heritage sites globally due to rising sea levels and more frequent disasters. It emphasizes the urgent need for innovative strategies to understand, preserve, and adapt these historic locations to safeguard them for future generations. Journal: *Nature*
- **Daily briefing: Human detritus remakes geology** — score: 0.000 The article discusses the evolving understanding of rocks in the context of human impact on geology, highlighting how human activities are reshaping the Earth's geological materials. It also touches on advancements in stem-cell research for severe autoimmune diseases and raises concerns about the phenomenon of 'AI deskilling,' where reliance on artificial intelligence may diminish human skills. Journal: *Nature*
- **Synergistically Facilitate Charge Dynamics and Suppress Energetic Disorder via Tailoring Solid Additive Substituents for Efficient Organic Solar Cells Over 21%** — score: 0.000 The article discusses advancements in optimizing organic solar cells (OSCs) by tailoring solid additive substituents, which enhance charge dynamics while simultaneously reducing energy loss. This dual approach aims to improve the overall efficiency of OSCs, achieving performance levels exceeding 21%. Journal: *RSC - Energy Environ. Sci. latest articles*
- **Charting the path to net-zero via CCS and urban-industrial system integration: A Swiss case study** — score: 0.000 This study explores the decarbonization potential of urban-industrial clusters in Switzerland's Rheintal valley by modeling the integration of energy and carbon systems.

It identifies cost-effective pathways to achieve a net-zero energy system by 2050, emphasizing the importance of coordinated sector planning to enhance efficiency and minimize dependence on fossil fuels, ultimately offering a scalable framework for regional decarbonization efforts. Journal: *Joule*

- **An alkyl phase-change medium refines morphology for 20.75% efficient organic solar cells** — score: 0.000 The article presents a novel phase-change medium used in the manufacturing of organic solar cells, which enhances the internal morphology of the films by allowing a brief rearrangement during processing. This method achieves an efficiency of 20.75% while demonstrating improved operational stability compared to traditional high-boiling-point additives, without the drawbacks of chemical residues. Journal: *Joule*